DAY-2

Probability (very) basic Definitions

DAV-2

→ Probability, Probability Distribution

→ Statistics.

↳ Descriptive Statistics

↳ Inferential Statistics.

DAV-3

Hypothesis Testing

Maths for DS - 2 months.

Disclaimer

→ Pace will be slower

→ Content will be 1.5 hrs - 1.75 hrs.

last time you used Probability?

12
4
8
6 Prerna ~5 years.
5
10
2 months - Project
10
1 m - Vibha

Where it can be used ?

↑

Data Science

"Likely"

Understanding chances of an event

Future = Uncertain ⟶ Decision Making.

↓

Predict? Minimize the future risk

Calculated.

AMAZON

UI/UX Designer
Product Designer

Suggestion?
What does the team think.?

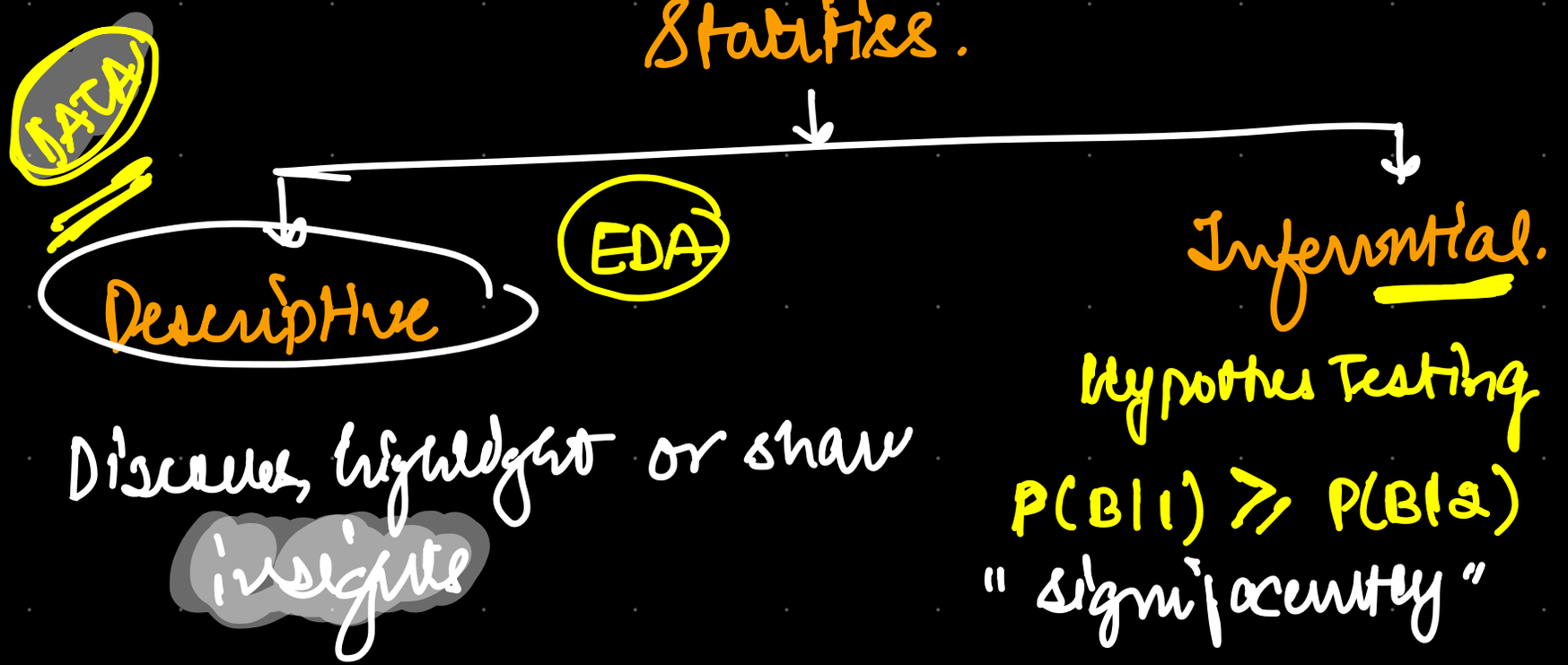
P(Buying given born)
P(Buying given just buy)

A/B Testing → Percent (A)
→ Format (B)



Probability – to model chances in terms of a number.

P, S, HT



Discover, highlight or share insights

Netflix

Popular on Netflix
Trending Top 10.

→ "Personalisation"

↙ Different for different customers.

P(Person will watch the content | → They watched X, Y, Z ...)

Top-10 movies with highest probability.

Probability

Coin toss, dice roll

Sample Sample

Outcomes

Favourable outcomes.

Experiment - Rolling a die.

Outcomes - $\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}$

Sample space - $\{1, 2, 3, 4, 5, 6\}$ — set.

Event - Indicent whose possible outcomes are **subset** of SS

$A = \{2, 4, 6\} \subset \{1, 2, 3, 4, 5, 6\}$.

Getting an even number on die

B - Getting a # smaller than 4

$\{1, 2, 3\} \subseteq SS$

↳ subset.

C - {1,3,5,7} Valid ??

{7} belong to SS.

Experiment - Tossing a coin

Sample Space - {H, T}

Events

{H} Getting a head

{T} Getting a tail

{H, T} Getting "something" on coin.

≠

HT ✗

→ either H or T

{ } Empty set is a valid event.

Experiment - Tossing 2 coins.

Sample Space - $\{HH, HT, TH, TT\}$

Event -

$A = \{HH, HT, TH\}$?

Atleast one head.

$B = \{HH, TT\}$

Getting same face on both coins.

Experiment - Tossing a coin → Rolling a die.

S = (Sample space) $= \{ H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6 \}$

Events ⇒ Coin is heads, $Q = \{H1, H2 \ldots H6\}$

$\#$

Dice is 3 $\{H3, T3\}$

Set Operations

Dice

$S = \{1, 2, 3, 4, 5, 6\}$

$A = \{1, 3, 5\}$

$B = \{1, 5, 6\}$ ✓

$A \cap B = \{1, 5\} = \{5, 1\} = B \cap A$

↳ common outcome of A and B.

$A \cup B = \{1, 3, 5, 1, 5, 6\} \rightarrow \{1, 3, 5, 6\}$.

↳ Any outcome of A or B

$= \{1, 3, 5\} + \{1, 5, 6\} - \{1, 5\}$

$A \cup B = A + B - A \cap B$

A^c → Complement = {2, 4, 6}

↳ Any valid outcome except A

Outcome in S that are not in A.

Experiment - Tossing a coin.

$S = \{H, T\}$

Chances of H - 50% — if coin is FAIR

Chances of T - 50%

Experiment - Rolling a fair die

$S = \{1, 2, 3, 4, 5, 6\}$

$A = \{2, 4, 6\}$

$B = \{1, 2, 3\}$

which one will you bet?

Prefer to bet. Why?

$$P(E) = \frac{\#\text{ favorable outcomes}}{\#\text{ total outcomes}}$$

$P(A) = 3/6 \rightarrow 0.5 \quad 50\%$

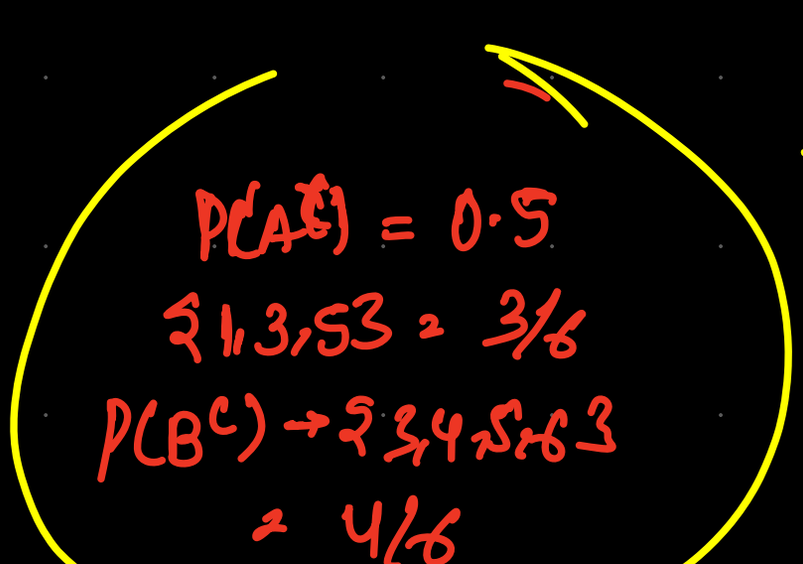
$P(B) = 2/6 \rightarrow 0.33 \quad 33\%$

$P(A^c) = 0.5$

$\{1, 3, 5\} = 3/6$

$P(B^c) \rightarrow \{3, 4, 5, 6\}$

$= 4/6$



$P(A) = 1 - P(A^C)$

$\frac{x}{N}$ $\quad$ $\frac{N-x}{N}$

$\frac{N}{N} - \frac{x}{N}$

$1 - \frac{x}{N}$

$P(A \cap B) = P(\{2,3\}) = 1/6$ → Repeative part

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$= P(\{1,3,5\}) + P(\{1,2\}) - P(A \cap B)$

$= 3/6 + 2/6 - 1/6 \rightarrow 4/6$

$A = \{2, 4, 6\}$ Roman

$B = \{1, 2\}$ Win.

$C = \{1, 3, 5\}$ Avant

Topic Single

$P(A \cap C) = P(\{\}) = 0$

Mutually Exclusive Event:

Disjoint Event - nothing in common



P(Attendance | VD)